

# Cephalometric landmark detection by considering translational invariance in the two-stage framework

Tao He<sup>a</sup>, Jie Yao<sup>b,c</sup>, Weidong Tian<sup>b</sup>, Zhang Yi<sup>a</sup>, Wei Tang<sup>b,\*</sup>, Jixiang Guo<sup>a,\*</sup>

<sup>a</sup> Machine Intelligence Laboratory, College of Computer Science, Sichuan University, Chengdu 610065, PR China

<sup>b</sup> Department of Oral and Maxillofacial Surgery, West China Hospital of Stomatology, Chengdu 610041, PR China

<sup>c</sup> College of Stomatology, Xian Jiaotong University, Xian, PR China

## ARTICLE INFO

### Article history:

Received 11 March 2021

Revised 15 June 2021

Accepted 9 August 2021

Available online 14 August 2021

### Keywords:

Deep convolutional neural networks

Translational invariance

Cephalometric landmark detection

## ABSTRACT

Recently, intelligent cephalometric landmark detection systems that employ deep convolutional neural networks (DCNNs) have been widely developed. In this paper, we concentrate on two challenging difficulties of DCNNs based cephalometric landmark detection. First, previous DCNN methods usually decreased the resolution of high-resolution cephalometric images. Hence, the trained DCNNs missed the detailed local features of original images. Moreover, the detected landmarks' locations were recovered using a fixed ratio, so location error was increased after recovery. Second, the performance of cephalometric landmark detection is dependent on DCNNs' translational invariance because the target landmarks' locations are influenced by any shift of the cephalometric images. However, modern DCNNs have limited translational invariance, and previous DCNN methods did not take translational invariance into consideration. In this paper, we developed the widely used two-stage framework to resolve the above two challenges. In the first stage, we train a global detection network, which receives the whole images as input, to generate candidate landmarks. In the second stage, we split the input image into patches and train a local refine network, which receives image patches as input, to refine the landmarks' locations. The proposed local refine network can produce accurate landmarks' locations because it entirely focuses on the detailed local features of high-resolution cephalometric images. The global detection network and local refine network are structured with translational invariance as their primary consideration. The proposed method achieved robust performance on a private dataset and the public Automatic Cephalometric X-ray Landmark Detection Challenge 2015 dataset.

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## 1. Introduction

Recently, with the development of artificial intelligence, intelligent cephalometric landmark detection (CLD) systems have been built to assist clinical orthodontists. Medical cephalometric images can be analyzed effectively, and the landmarks within them can be detected precisely with the help of deep convolutional neural networks (DCNNs). However, clinical cephalometric measurements are sensitively influenced by pixel-level errors of landmarks' locations, and intelligent CLD systems require highly accurate detection of landmarks' locations [1]. Therefore, intelligent CLD systems usually have human-computer interaction options to correct the landmarks' locations. Although CLD has

achieved great success through combination with DCNN methods [2–8], it is still a challenge to improve the accuracy of CLD.

In this paper, we concentrate on two difficulties surrounding the use of CLD with DCNN methods. We call the first problem as *high-resolution loss* because the source cephalometric images usually have high resolution. For example, the resolution of the images on the public Automatic Cephalometric X-ray Landmark Detection Challenge (ACXLDC) 2015 dataset<sup>1</sup> is  $1935 \times 2400$ . We also collected a private dataset from the West China Hospital of Stomatology, which consists of images with the resolution of  $1752 \times 1537$ , and named it the Huaxi-Analysis dataset. Previous DCNN methods have usually decreased the resolution of high-resolution cephalometric images to reduce the computational complexity, thus the trained DCNNs missed source images' detailed local features. However, the detailed local features are critical because the aim of CLD is specifying landmarks' locations, which is a pixel-level prediction

\* Corresponding authors.

E-mail addresses: [mydrtw@vip.sina.com](mailto:mydrtw@vip.sina.com) (W. Tang), [guojixiang@scu.edu.cn](mailto:guojixiang@scu.edu.cn) (J. Guo).

<sup>1</sup> <http://www-o.ntust.edu.tw/cweiwang/ISBI2015/challenge1/>.

task. The location of a target landmark pixel is usually highly relative to those of its neighboring pixels. Moreover, *high-resolution loss* tends to induce recovery error. Chen et al. [2] reduced the resolution of images from the ACXLDC 2015 dataset to  $640 \times 800$  (scale ratio: 0.3) and fed them into a DCNN. A landmark with a mean radial error (MRE) of 1 mm (mm) would have a corresponding MRE of 3.33 mm after recovery with the fixed scale ratio. However, it is impractical to directly feed high-resolution images into a DCNN because it wastes computational resources and is very time-consuming.

We call the second problem the *translational invariance limit*. Specifying a landmark's location can be viewed as specifying the pixel shift to the original dot, so that the essential task of CLD is to predict pixel shifts. A good CLD method should have stable and robust performance to image pixel shift. However, modern DCNNs have essential limitations in terms of translational invariance. Richard Zhang [9] pointed out that the reason for these limitations is that commonly used downsampling methods (e.g., max-pooling, stride convolution, and average-pooling operations) ignore the sampling theorem. The anti-aliasing method by low-pass filtering followed by downsampling harms DCNNs' performance. An anti-aliasing module that makes DCNN shift-invariant to some degree was proposed by Richard Zhang [9]. Azulay and Weiss [10] pointed out that neither the convolutional architecture nor data augmentation is sufficient to solve this problem, which they addressed it by anti-aliasing the network's intermediate representations and increasing data augmentation. However, they concluded that the problem of ensuring that neural networks are invariance to small image transformations remains unsolved. In Fig. 1 (b), we illustrate the MREs with respect to the impact of image translation performed by a baseline DCNN structured based on Chen et al. [2] but without the attentive feature pyramid fusion module. The MREs are drastically changed when the image is shifted within a 5-pixel range. As far as we know, previous DCNN structures did not take the *translational invariance limit* problem into account for CLD.

To solve the *high-resolution loss* and *translational invariance limit* problems, we develop the two-stage CLD framework. In the first stage, we train a *global detection network (GDN)*, which receives the whole images as input, to generate candidate landmarks. In the second stage, we split the input image into patches and train a *local refine-network (LRN)*, which receives image patches as input, to refine the landmarks' locations. The LRN can produce accurate landmarks' locations because it exclusively focuses on the detailed local features of high-resolution cephalometric images. This type of two-stage framework, also called a

coarse-to-fine framework, has been widely used in medical imaging tasks, such as medical image segmentation [11,12] and nodule detection [13,14]. Zhong et al. [3] was the first to propose the two-stage framework for CLD, and it achieved the state-of-the-art results on the public ACXLDC 2015 dataset. In this paper, we develop the two-stage framework further by considering DCNN's translational invariance. The proposed framework is illustrated in Fig. 2, and the following are its contributions:

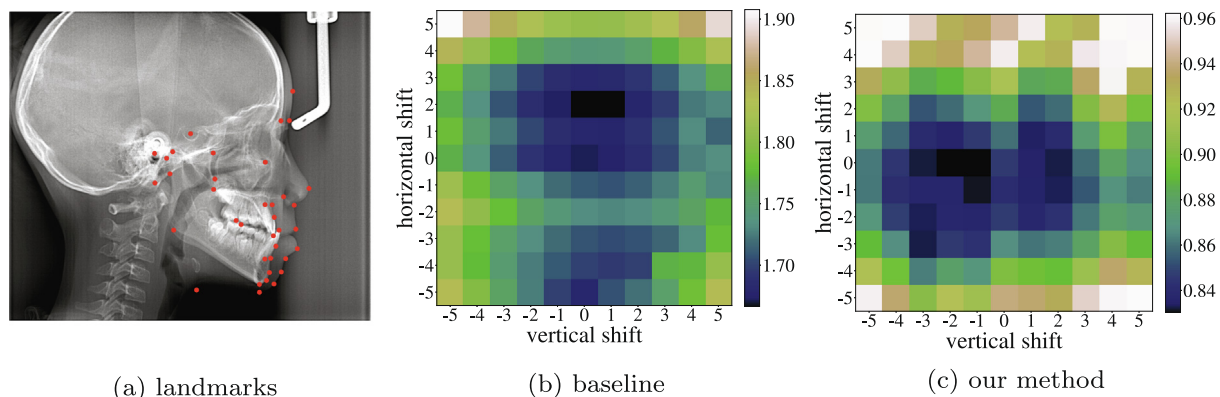
- We structure the GDN and LRN with translational invariance as the primary consideration. First, we reduce the number of downsampling layers in the LRN; second, we replace the general pooling layers of the GDN and LRN with *anti-aliased pooling layers*; and third, we implement a *blurring upsampling module* in the GDN, which shifts the feature maps and averages the upsampled results.
- We implement the multi-task strategy to improve the GDN's generalization ability by jointly performing the *offset map regression* and *Gaussian heatmap regression*. We also reconstruct the atrous spatial pyramid pooling (ASPP) module to achieve better performance.
- The proposed method has robust translational invariance and achieves outstanding performance on the private Huaxi-Analysis and public ACXLDC 2015 datasets. The example in Fig. 1 (c) illustrates that our method has more robust performance than the baseline method.

The rest of this paper is structured as follows: in Section 2, we introduce related work about traditional methods, deep learning methods, and three-dimensional (3D) CLD; in Section 3, we present the proposed two-stage framework; in Section 4, the experimental results are given; and in Section 5, we conclude the work and discuss future work.

## 2. Related work

### 2.1. Traditional methods

The international symposium on biomedical imaging conference has held automatic cephalometric landmark detection challenges twice in 2014 [15] and 2015 [1]. Most methods that had outstanding performance in the challenges were traditional machine learning methods that did not involve deep learning. Random Forest (RF) was widely used by most of the challenge partic-



**Fig. 1.** (a) indicates a cephalometric image from the West-Analysis dataset that contains 37 landmarks. (b) and (c) plot the mean radial errors (MREs) with respect to the impact of vertical and horizontal pixel shifts. Because the dimension of one pixel on the West-Analysis dataset is 0.13 mm, the image shift ranges from  $-0.65$  mm to  $0.65$  mm. The baseline method changes the MRE from 1.67 mm to 1.91 mm ( $\pm 0.24$  mm), and the proposed method changes the MRE from 0.83 mm to 0.96 mm ( $\pm 0.13$  mm).

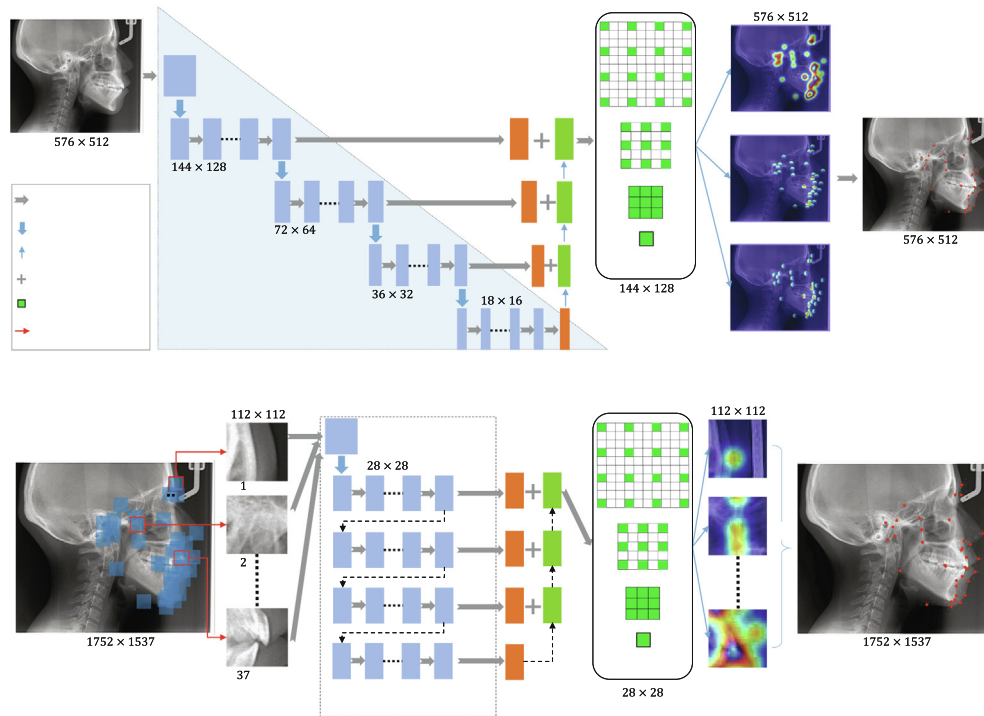


Fig. 2. The two-stage framework.

ipants. Ibragimov et al. [16] used RF to create a candidate landmark point detector by combining a set of landmarks' Haar-like features. Image pixels with maximal detector response indicated the landmark candidate points. The framework relied on spatial relationships among landmarks to select the final candidate points. RF has also been used to model the spatial relationships between subsets of landmarks. Ibragimov et al. [17] optimized their landmark detection algorithm by game theory with shape and appearance-based landmark refinement. The appearance-based landmark refinement combined the global and local intensity information for more accurate detection. Chu et al. [18] proposed a two-stage framework, where RF was used for the training step in the first stage. They trained 19 RF landmark detectors individually, and in the second stage, the prior shape constraints were used to modify the landmark positions. Mirzaalian et al. [19] used RF to compute a likelihood for each landmark on a set of extracted features. Then, they used a pictorial structure algorithm to globally optimize the cost function with a pairwise regularization term. These challenge participants obtained high competitive grades using these RF methods.

Other related methods have also contributed to the performance of CLD systems. Lindner and Cootes [20] trained multiple RFs to vote for candidate landmark locations, and optimized the total votes overall landmark locations based on the regression-voting. They also applied this method in a constrained local model framework as a part of a fully automatic landmark detection system [21]. A statistical shape model is matched to the landmark predictions to ensure consistency. Vandaele et al. [22] proposed a tree-based algorithm, named extremely randomized forests, combined with simple pixel-based multi-resolution features. To solve the unbalanced learning problem, they randomly sampled partial points for training in the images and learned observations on surrounding of 40 mm around each landmark. Cheng et al. [23] used a data-driven method that jointly estimated the displacements of all patches to estimate landmark displacement from image patches.

They conceptualized displacement estimation as a convex optimization problem and considered the geometric constraints on the test set.

## 2.2. Deep learning methods

In recent years, the use of DCNNs in CLD systems has been widely developed. Different from traditional methods, DCNNs are trained on source cephalometric images but not on hand-crafted features. Christian et al. [4] used DCNN-based heatmap regression for multiple landmark localization. They performed end-to-end training of a single DCNN by regressing heatmaps instead of using a sliding window approach to generate multiple landmark candidates. Heatmap regression was also widely used in related tasks, such as mitosis detection [24]. Christian et al. [5] also evaluated their method on CLD and proved that heatmap regression is an efficient CLD method. They optimized the method by integrating spatial configuration into heatmap regression. The proposed fully convolutional SpatialConfiguration-Net (SCN) in which one component produced candidate landmarks and the other component improved robustness by incorporating the spatial configuration of landmarks. Arik et al. [6] directly regressed landmark locations using DCNNs and yield quantitative estimation of pathologies in the jaws and skull base regions. In this paper, we also use DCNNs with the heatmap regression.

Recent work has built specific neural network modules that configure DCNNs to improve the performance of CLD methods. Chen et al. [2] proposed a DCNN with an attentive feature pyramid fusion module (AFPF) to semantically enhanced the fusion features and explicitly shape high-resolution. They regressed the offset maps as multiple votes and used pixel-wise voting to select the most likely landmark locations. Park et al. [7] applied two object detection methods, You-Only-Look-Once version 3 and single shot multibox detector, to a CLD task on their private dataset with 80 landmarks. Qian et al. [8] proposed a Faster R-CNN based method,

named CephaNet, for cephalometric landmark detection. Multi-task learning and multi-scale training strategies were used to reduce intra-class variations and detect small landmarks. They also proposed a method to remove abnormal detected landmarks. In this paper, we build network modules by considering translational invariance.

### 2.3. Three-dimensional cephalometric landmark detection

Landmark detection of cephalometric images is not the only method by which it is possible to perform CLD tasks. Some work has focused on 3D CLD by analyzing cone-beam computed tomography images. Zhang et al. [25] proposed a segmentation-guided partially-joint regression model that combined a regression voting strategy with multi-scale statistical features. The regression voting strategy localized landmarks by aggregating evidence from context locations relieving the problems caused by image artefacts. High-level multi-scale statistical features were extracted by a fast quantization method to describe each voxel's appearance. Ibragimov et al. [26] developed a game theory-based method that applied three different shape representations based on transportation theory. Landmarks were considered as destinations and sources, connections between all possible landmarks were considered as roads, and established landmark connections were considered as goods transported via the roads.

DCNNs have also been applied in 3D CLD. Context-guided fully convolutional networks for joint craniomaxillofacial bone segmentation and landmark digitization were proposed by Zhang et al. [27]. The multi-task learning strategy was used because the bone segmentation and landmark digitization tasks were strongly associated. The highlight of that method was the use of a displacement map, which modeled information about spatial context. Torosdagli et al. [28] proposed a three-step framework for segmentation and anatomical landmark detection. They first used a DCNN to perform image segmentation, and then the landmark localization of sparsely-spaced anatomical landmarks was formulated in geodesic space. Finally, a long short-term memory was applied to identify closely spaced landmarks. However, CLD on two-dimensional cephalometric images is still the most common method for orthognathic measurement and maxillofacial surgery. In this paper, we do not employ the 3D CLD domain.

## 3. Method

The two-stage framework (illustrated in Fig. 2) was proposed with the goal of solving the *high-resolution loss* problem. In the global stage, a deep GDN is constructed to generate candidate landmarks. First, the cephalometric image is zoomed out to an appropriate image resolution and fed into a deep fully connected network (FCN), which adopts the ResNet architecture [29] as its network backbone. Then, the feature maps produced by the FCN are input to an ASPP module. Finally, Gaussian heatmaps and offset maps are combined to form the cost function.

In the local stage, a deep LRN is constructed to generate accurate landmarks. Before network training, the whole cephalometric image is split into patches by randomly selecting the center of each patch from the area adjacent to each landmark. Then, all of the selected small patches are fed into a light version of the FCN followed by an ASPP module. Finally, all of the Gaussian heatmaps are merged, and prediction is run to localize landmarks accurately. We do not apply offset maps in the LRN because calculating the offset maps of all patches would greatly increase the computational complexity of the CLD task.

The inference stage takes a new cephalometric image as input. First, the image is zoomed out, and the GDN is used to generate the

heatmaps and corresponding offset maps. The heatmaps and offset maps are then used to create two different groups of landmarks by regression-voting. The candidate landmarks based on the Gaussian heatmaps were the mean votes for landmarks within areas that had predicted probabilities larger than 0.5. Furthermore, the candidate landmarks based on the offset maps were located at the positions with the largest numbers of votes among those within the concerned areas: we used the positions with the largest numbers of votes as the candidate landmarks. **We averaged the locations obtained by these two methods as the final locations of the candidate landmarks.** Second, we split the image into patches according to these candidate landmarks and used the LRN's heatmaps to produce accurate landmarks' locations. The LRN has limited ability to fine-tune the locations of candidate landmarks because some candidate landmarks are far away from the ground truth locations. Thus, we used **expansive exploration** [3] as a post-processing method to boost the performance of this refining process.

**Translational Invariance Strategies.** To solve the *translational invariance limit* problem, we modified three elements of the two-stage framework. First, small-sized image patches were used as the inputs to the LRN, giving the FCN a light structure. Because the downsampling layers of DCNNs destroy the networks' translational invariance, we pruned three downsampling layers from the backbone network of LRN, and its computation was still within the acceptable range of complexity. Second, we replaced the rest downsampling layers of the GDN and LRN with anti-aliased pooling layers, as anti-aliased pooling layers have shown robust performance in image recognition tasks [9]. Third, as the anti-aliased pooling layers cannot deal with the situation of upsampling, we propose a new upsampling method named average upsample that is based on the strategy of blurring feature maps.

**Anti-aliased Pooling and Average Upsample Layers.** The anti-aliased pooling and average upsample layers are illustrated in Fig. 3. The anti-aliased pooling layer consists of a max function, an anti-aliasing filter, and downsampling, where the max function and anti-aliasing filter preserve shift-equivariance. The average upsample blurs the features by averaging neighborhood features. First, the feature map is shifted with padding, and then the average upsample results are taken as the final activation. Fig. 4 illustrates the benefits of average upsample using the same example as the one shown by Richard Zhang [9]. The average upsample layers have closer internal representations than the general upsample layers because of the averaging of the shifted feature maps. We designed two types of averaging strategies: *cruciform* and *full*. In the *cruciform* type, we averaged five shifted feature maps including one-pixel shifts to the left, right, above, and below. In the *full* type, we averaged nine shifted feature maps, including diagonal shifts to the other four corners. Fig. 5 illustrates the comparison between the general upsample and the proposed average upsample.

### 3.1. Network training

Given a cephalometric image  $X$ , CLD aims to detect anatomical landmarks  $L = (L_1, L_2, \dots, L_n)$ , where  $L_n$  denotes the  $n$ -th landmark's location. A backbone network was used as the feature extractor to process the image, as follows:

$$F_1, F_2, F_3, F_4 = \text{net}(X). \quad (1)$$

we obtained four feature maps with different scales, in which ResNets were used as the network backbones. In the GDN, we replaced all of the downsampling layers with anti-aliased pooling layers, and in the LRN, we used the anti-aliased pooling layer as the first downsampling layer and directly pruned the remaining three downsampling layers. Then, we applied a  $1 \times 1$  convolution on these feature maps to obtain four feature maps (orange rectan-

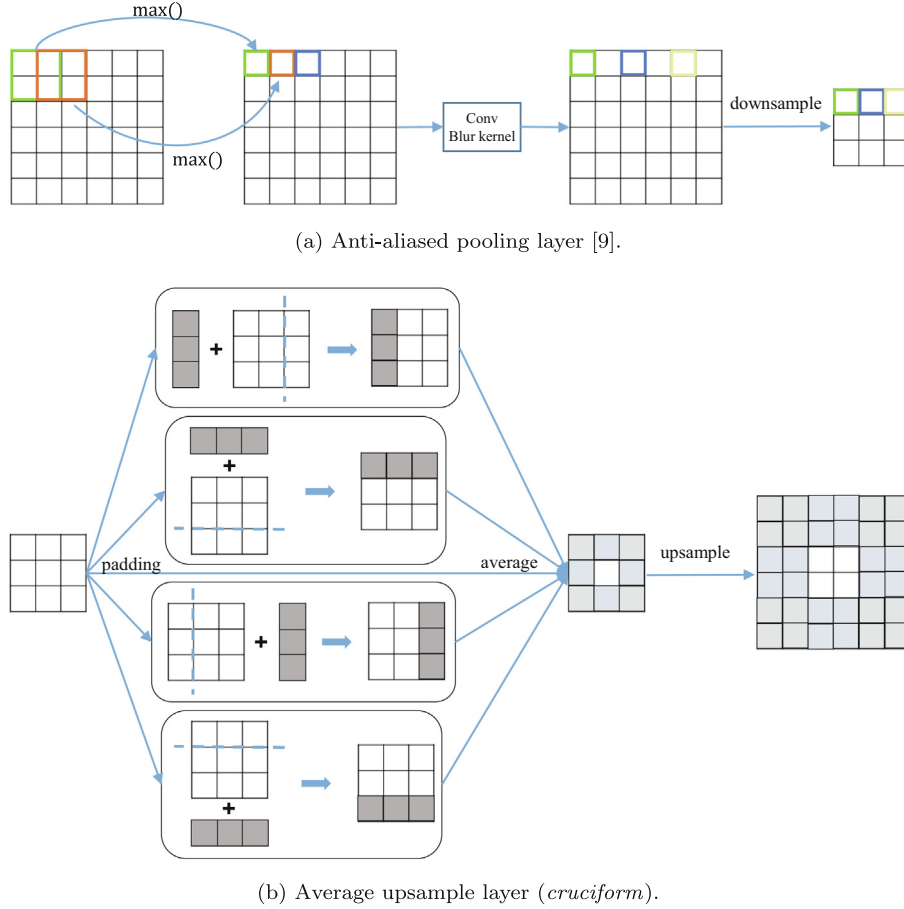
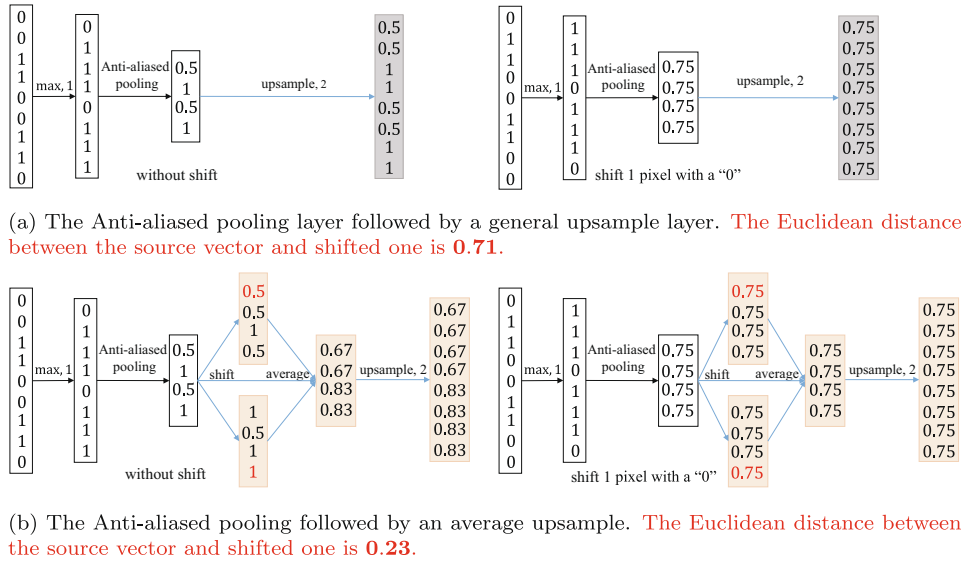


Fig. 3. The anti-aliased pooling and average upsample layers.

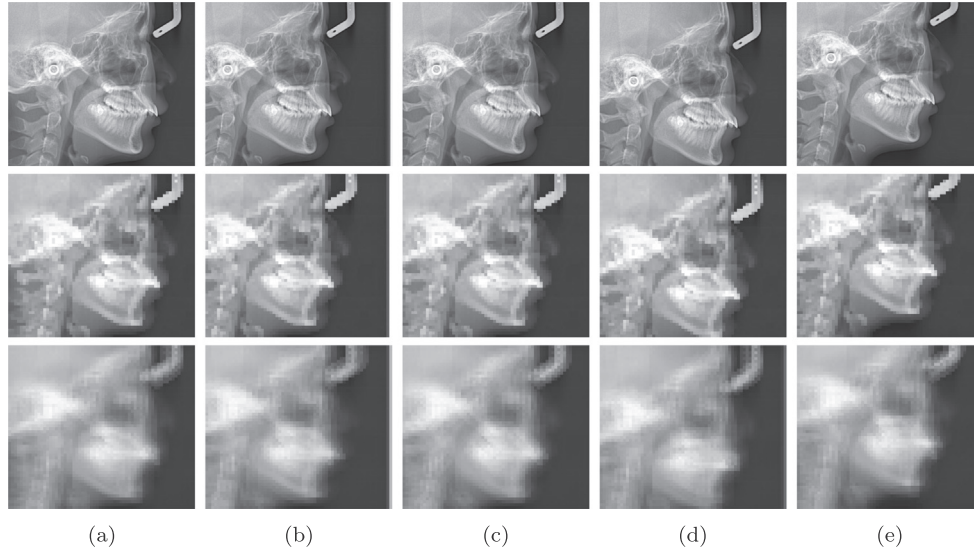


**Fig. 4.** Anti-aliased pooling and average upsample layers. The neighborhood pixels for the average upsample were unknown. We used the copies of pixels in the border as the padded pixels (“0.5” and “1” in red color). As we can see that  $(d_a) = 0.71 > (d_b) = 0.23$ , it means that the average upsample layers produce closer internal representations than the general upsample layers. Thus, the pixel shift induces fewer feature changes by replacing the general upsample layers with the average upsample layers.

gles in Fig. 2) with 64, 128, 256, and 512 adjustable channels, as follows:

$$E_i = conv_{1 \times 1}(F_i), \quad (2)$$

where  $i = 1, 2, 3$ , and 4, respectively. In the LRN, we directly concatenated these feature maps. In the GDN, the upsampling feature maps (green rectangles in Fig. 2) were generated by the proposed average upsample layers as follows:



**Fig. 5.** The examples of comparison between the general upsample (the second row) and the proposed average upsample (the third row). The first row includes source images. We directly use three downsample and upsample layers to process the images. (a), (b), (c), (d), and (e) indicates shifting to the right, shifting to the left, no shift, shifting to the bottom, and shifting to the top, respectively. The average upsample layers are less affected by the shifting than general upsample layers.

$$U_{j-1} = \text{upsample}(U_j), \quad (3)$$

where  $U_4 = E_4$  and  $j = 2, 3$ , or  $4$ . The upsampling was performed using bilinear interpolation. Then, we concatenated  $E_i$  and  $U_j$  together step-by-step. We applied an ASPP module to extract the multi-scale image context and then used average upsample to obtain the final feature map  $C$ , the channel size of which was equal to the number of landmarks for LRN and three times the number of landmarks for GDN.

To build the target, we first set an image mask  $M$  with the same size as the input. We defined a circle with radius  $R$  inside the mask to indicate the concerned area, where inside and other pixels were assigned values of 1 and 0, respectively. We set  $R = 20$  empirically in this paper. In the LRN, we modeled the targets as Gaussian heatmaps:

$$\begin{cases} H_i^{jk} = M^{jk} \cdot \exp \left[ -\frac{(j-L_i^{(1)})^2 + (k-L_i^{(2)})^2}{\alpha \cdot 2\sigma^2} \right], \\ P_h = \frac{1}{nJK} \sum_{i=1}^n \sum_{j=1}^J \sum_{k=1}^K |C_i^{jk} - H_i^{jk}|. \end{cases} \quad (4)$$

where  $(j, k)$  denotes each position in the feature maps,  $(\cdot)$  denotes the pixel-wise multiplication,  $P_h$  is the cost function, and  $L_i^{(1)}, L_i^{(2)}$  denotes the row and column of the landmark's location, respectively. We set  $\sigma = R$  and  $\alpha = 0.1$  to control the size of Gaussian distribution.

In the GDN, we adopted not only Gaussian heatmap regression but also offset map regression, which was inspired by Chen et al. [2]. The formulation is as follows:

$$\begin{cases} O_i^j = M^{jk} \cdot \left( \frac{L_i^{(1)} - j}{R} \right), \\ O_i^k = M^{jk} \cdot \left( \frac{L_i^{(2)} - k}{R} \right), \\ P_o = \frac{1}{nJK} \sum_{i=1}^n \sum_{j=1}^J \sum_{k=1}^K (|C_{n+i}^{jk} - O_i^j| + |C_{2n+i}^{jk} - O_i^k|), \\ P = \gamma P_h + P_o. \end{cases} \quad (5)$$

where  $\gamma$  is used to balance the loss function terms, and  $P$  is the GDN's multi-task loss function.  $\gamma$  was set as 0.5 empirically.

## 4. Experiment

In this section, we first introduce the private Huaxi-Analysis and public ACXLDC datasets. Then, we describe several ablation studies with respect to the ASPP module, downsampling, anti-aliased pooling, average upsample, Gaussian heatmap, and offset map. Finally, we provide detailed experimental results and analysis.

### 4.1. Datasets and experimental setup

The public ACXLDC dataset contains a total of 400 cephalometric images (150, 150, and 100 for training, validation, and testing, respectively). Each image contains 19 landmarks and has a resolution of  $1935 \times 2400$  pixels with pixel spacing of 0.1 mm. The Huaxi-Analysis dataset contains 1005 cephalometric images (605, 200, and 200 for training, validation, and testing, respectively). Each image contains 37 landmarks and has a resolution of  $1752 \times 1537$  pixels with pixel spacing of 0.13 mm.

The Huaxi-Analysis dataset was labeled by experienced orthodontists from the West China Hospital of Stomatology. Each cephalometric image was labeled in two stages. In the first stage, the orthodontists directly located the landmarks in the image. Two copies of some landmarks, such as porion, gonion, and orbitale, were visible in the image because of effects of the camera device. In this situation, the orthodontists labeled both copies of the landmark and averaged their locations as the final locations. In the second semi-automatic stage, we first trained a CLD model and built a preliminary version of the CLD system for producing pre-located landmarks, the video summary of which is available at YouTube.<sup>2</sup> Then, the orthodontists relocated these landmarks with the assistance of our CLD system. Specifically, in the first stage, 512 cephalometric images were manually located by orthodontists. Each sample took average half an hour to locate. To speed up the labeling process, we train a CLD model using these 512 images as the backend of the CLD system. Based on the pre-located landmarks, orthodontists just spend average 5 min modifying the location of rest unlabeled images.

<sup>2</sup> [https://www.youtube.com/watch?v=gouyyoj\\_mb4](https://www.youtube.com/watch?v=gouyyoj_mb4).

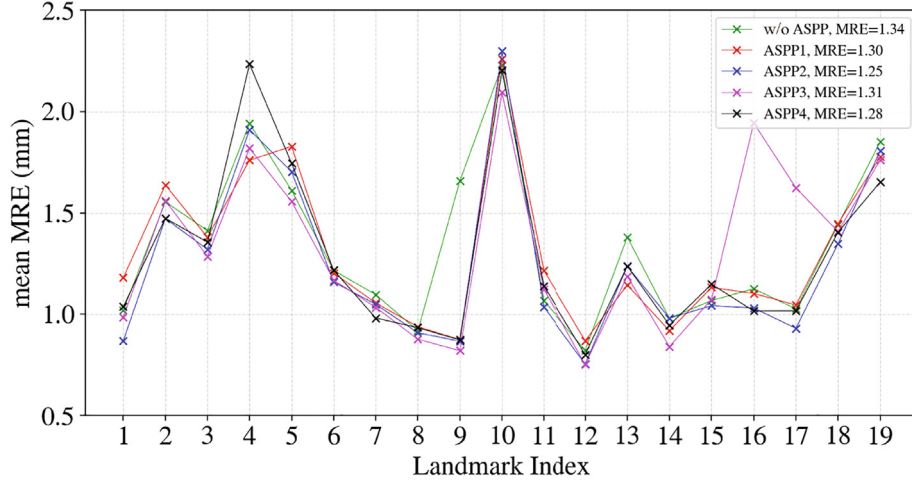


Fig. 6. The experimental results with different configurations of dilation factors in ASPP. w/o ASPP means “without ASPP”.

Table 1

The detailed configurations of ASPP. w/o ASPP means “without ASPP”.

| name     | dilation rates ( $r$ ) |
|----------|------------------------|
| w/o ASPP | [1]                    |
| ASPP4    | [1, 6, 12, 18]         |
| ASPP3    | [1, 12, 24, 36]        |
| ASPP2    | [1, 3, 7, 13, 20]      |
| ASPP1    | [1, 2, 5, 9, 14, 20]   |

We perform all ablation experiments on the public ACXLDC dataset and comparative experiments on both datasets. All experiments in this section were implemented with PyTorch [30] using an NVIDIA Titan XP GPU. We used stochastic gradient descent with momentum 0.9 for training. The learning rate was initialized as 0.1 and decayed exponentially by the factor of 0.98. The weight decay value was  $5e^{-4}$ . For data augmentation, we adopted random cropping by a scaling factor of between 0.6 and 1 for the GDN and cropped the image patches by a scaling factor ranging between 0.9 and 1.1 for the LRN. The center of the image patch was selected within a circle of a radius 20 pixels. To deal with the unbalanced

‘training samples, we performed hard negative mining to choose a fixed number of negative samples outside the heatmap circles for training. In the global stage, we reduced the image resolutions of the two datasets to  $640 \times 800$  and  $576 \times 512$ , respectively.

MRE with standard deviation (Std) and the successful detection rate (SDR) were used as the evaluation metrics. They are defined as follows:

$$\left\{ \begin{array}{l} MRE = \frac{\sum_{i=1}^n R_i}{n}, \\ Std = \sqrt{\frac{\sum_{i=1}^n (R_i - MRE)^2}{n-1}}, \\ SDR = \frac{\text{number of accurate detection}}{\text{number of detection}} \times 100\%. \end{array} \right. \quad (6)$$

where  $n$  denotes the number of images and  $R_i$  denotes the mean Euclidean distance between the ground truth and predicted locations in the  $i$ th image.

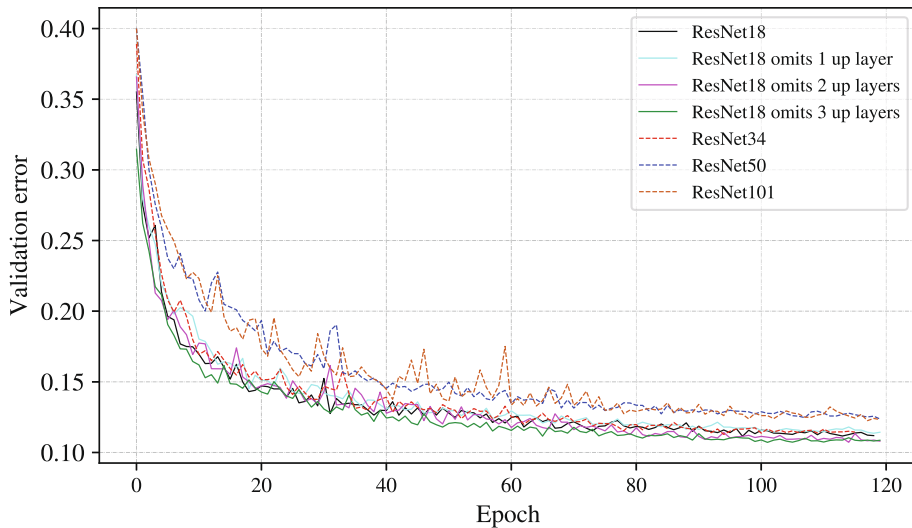


Fig. 7. The validation errors with different ResNets. “up” indicates the upsample layer.

**Table 2**

The experimental results of GDN with different network modules on ACXLDC validation set.

| No. | Gauss heatmap | Offset map | Anti-aliased pooling | Average upsample   | MRE $\pm$ Std (mm)                | SDR (%)      |              |              |              |
|-----|---------------|------------|----------------------|--------------------|-----------------------------------|--------------|--------------|--------------|--------------|
|     |               |            |                      |                    |                                   | 2 mm         | 2.5 mm       | 3 mm         | 4 mm         |
| 1   | ✓             |            |                      |                    | 1.32 $\pm$ 1.01                   | 82.53        | 87.96        | 92.21        | 96.18        |
| 2   |               | ✓          |                      |                    | 1.36 $\pm$ 0.99                   | 83.51        | 89.89        | 93.89        | 96.77        |
| 3   | ✓             | ✓          |                      |                    | 1.29 $\pm$ 1.08                   | 83.23        | 90.21        | 94.04        | 97.65        |
| 4   | ✓             | ✓          | ✓                    |                    | 1.26 $\pm$ 0.70                   | 84.07        | 90.35        | 93.93        | 97.23        |
| 5   | ✓             | ✓          |                      | ✓ <i>cruciform</i> | 1.18 $\pm$ 0.49                   | <b>86.77</b> | <b>91.93</b> | <b>94.49</b> | <b>97.89</b> |
| 6   | ✓             | ✓          | ✓                    | ✓ <i>cruciform</i> | <b>1.15 <math>\pm</math> 0.22</b> | 86.63        | 91.82        | <b>95.16</b> | <b>98.25</b> |
| 7   | ✓             | ✓          | ✓                    | ✓ <i>full</i>      | 1.17 $\pm$ 0.27                   | 85.47        | 91.44        | 94.66        | 97.96        |

**Table 3**

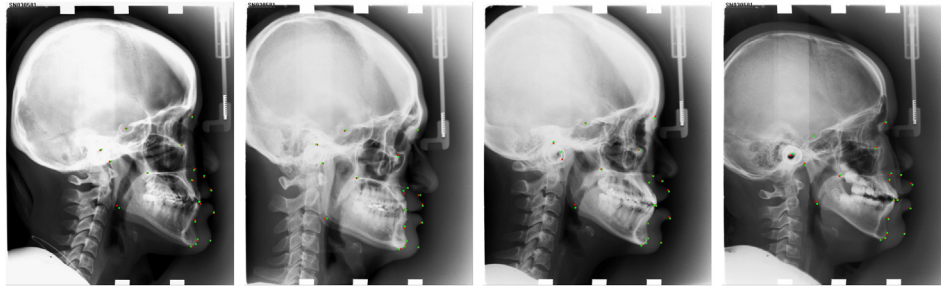
Comparison with other related methods on ACXLDC dataset, where Qian et al. [8] did not present the MRE and Std, and Chen et al. [2] and Yao et al. [31] did not present the Std.

| Method                  | MRE $\pm$ Std (mm) | SDR (%) |        |       |       |
|-------------------------|--------------------|---------|--------|-------|-------|
|                         |                    | 2 mm    | 2.5 mm | 3 mm  | 4 mm  |
| Validation set          |                    |         |        |       |       |
| Lindner and Cootes [20] | 1.67 $\pm$ 1.65    | 74.95   | 80.28  | 84.56 | 89.68 |
| Qian et al. [8]         |                    | 82.50   | 86.20  | 89.30 | 90.60 |
| Zhong et al. [3]        | 1.14 $\pm$ 1.03    | 86.74   | 92.00  | 94.71 | 97.82 |
| Chen et al. [2]         | 1.17               | 86.67   | 92.67  | 95.54 | 98.53 |
| Yao et al. [31]         | 1.24               | 84.84   | 90.52  | 93.75 | 97.40 |
| Oh et al. [32]          | 1.18 $\pm$ 1.01    | 86.20   | 91.20  | 94.40 | 97.70 |
| Ours                    | 1.09 $\pm$ 0.31    | 87.47   | 92.89  | 95.15 | 98.25 |
| Test set                |                    |         |        |       |       |
| Chen et al. [2]         | 1.48               | 75.05   | 82.84  | 88.53 | 95.05 |
| Yao et al. [31]         | 1.61               | 71.89   | 80.63  | 86.36 | 93.68 |
| Oh et al. [32]          | 1.46 $\pm$ 0.82    | 75.90   | 83.40  | 89.30 | 94.70 |
| Ours                    | 1.43 $\pm$ 0.22    | 75.47   | 83.44  | 87.84 | 94.63 |

**Table 4**

The experimental results on Huaxi-Analysis dataset.

| Set  | LRN | MRE $\pm$ Std (mm) | SDR (%) |        |       |       |
|------|-----|--------------------|---------|--------|-------|-------|
|      |     |                    | 2 mm    | 2.5 mm | 3 mm  | 4 mm  |
| Val  |     | 1.30 $\pm$ 0.28    | 89.42   | 93.51  | 96.14 | 98.42 |
|      | ✓   | 1.24 $\pm$ 0.26    | 89.50   | 93.88  | 95.92 | 98.41 |
| Test |     | 1.26 $\pm$ 0.20    | 89.78   | 93.92  | 96.20 | 98.65 |
|      | ✓   | 1.21 $\pm$ 0.19    | 90.30   | 94.59  | 95.95 | 98.61 |

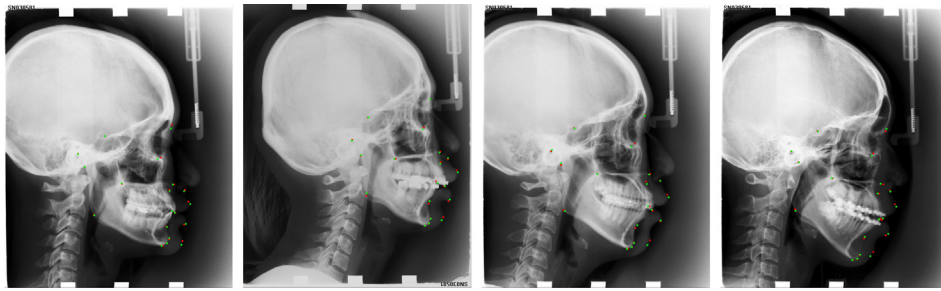
**Fig. 8.** The examples on ACXLDC validation set.

#### 4.2. Ablation studies

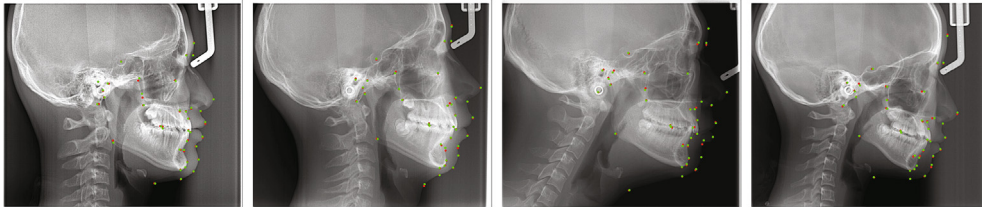
The ASPP module is important because it extracts multi-scale image context features and has been verified in semantic segmentation tasks [33]. One of ASPP's most important parameters is the dilation factor. Considering that cephalometric images have high resolution, we fine-tuned the dilation factor vector and trained a GDN to compare the MREs of the ACXLDC validation set. The experimental results for each landmark are plotted in Fig. 6 and the corresponding dilation factor vectors are listed in Table 1. ASPP2

achieved the best MRE, and ASPP3 achieved good results for the majority of landmarks but produced large MREs for landmarks 16 and 17. We guess that the high dilation rate ( $r = 36$ ) generates useless features convolved from the padding and that over-fitting may therefore take place when ASPP3 is used on these landmarks. We used ASPP2 in all remaining experiments.

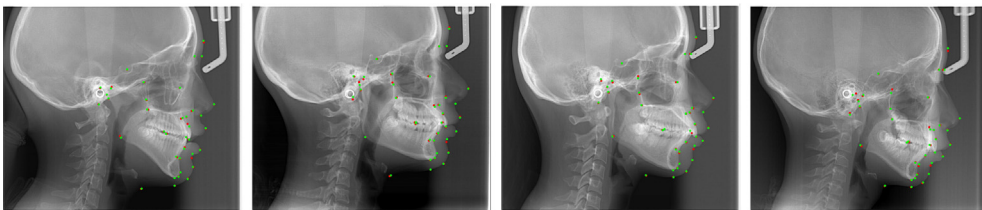
To validate that pruning the downsampling layers of the LRN is useful for addressing the *translational invariance limit* problem, we compared the validation error with that obtained by pruning the downsampling layers of ResNet18 backbone. In Fig. 7, we also plot-



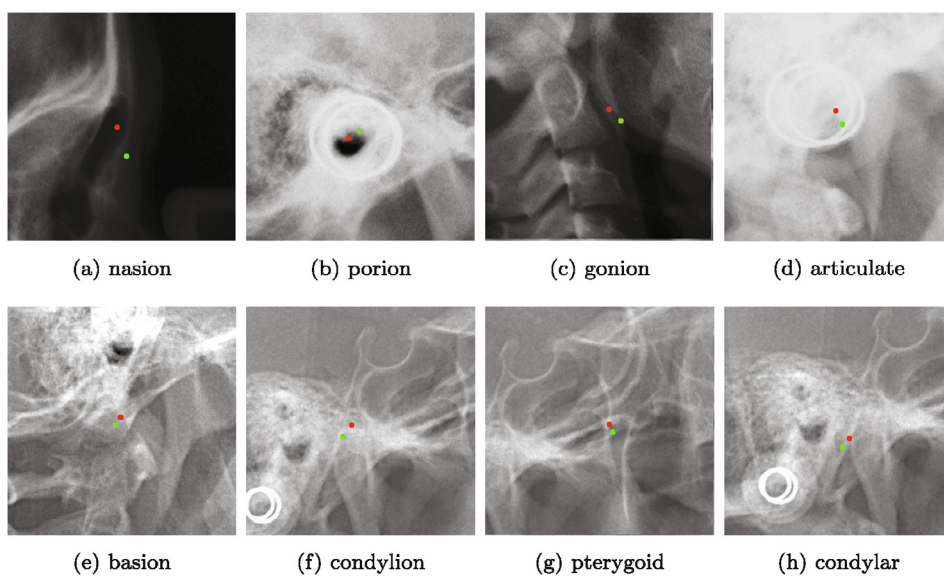
**Fig. 9.** The examples on ACXLDC test set.



**Fig. 10.** The examples on Huaxi-Analysis validation set.



**Fig. 11.** The examples on Huaxi-Analysis test set.



**Fig. 12.** The illustration of detected landmarks with large MREs.

ted the learning curves of ResNet34, ResNet50, and ResNet101, which employ pre-trained parameters. The results indicate that using deeper ResNets as the FCN's encoder does not produce fewer errors but that pruning more downsampling layers from a ResNet18 can improve performance.

To evaluate the anti-aliased pooling, average upsampling, Gaussian heatmaps, and offset maps, we trained a GDN with a ResNet18 backbone and compared its results on the ACXLDC validation set with those of the proposed model (Table 2). We set Binomial-5 as the blur kernel of anti-aliased pooling. Experiment No. 1, 2, and 3 indicate that multi-task learning improves the experimental results, and the results of experiment No. 4 and 5 indicate that both anti-aliased pooling and average upsample layers provide useful performance improvements. The results of Experiments No. 6 and 7 indicate that all of the modules used are helpful for the CLD tasks. The *cruciform* type has slightly better performance than the *full* type of average upsample. Thus, in the remaining experiments, we use *cruciform*-type average upsample.

#### 4.3. Experimental results and analysis

In this section, we compare the proposed method with other related methods and compare our method's experimental results with vs. without LRN. The experimental results are listed in Tables 3 and 4. Three experimental observations seem important. First, our method achieved better MRE results than the previous two-stage method [3] and other related methods. Our method has the best stable MRE, as it has the lowest Std. Second, the LRN enhances MRE and SDR within 2.5 mm because the LRN mainly refined candidate landmarks within 2.5 mm. Third, both our method and that of Chen et al. [2] resulted in overfitting on the ACXLDC dataset, but over-fitting was reduced by including a sufficient number of training images: the experimental results were outstanding on Huaxi-Analysis test set.

Some examples are given in Fig. 8–11, where the red dots indicate the groundtruth landmark locations and the green dots indicate the predicted landmark locations. A full correct hit presents that the green dot covers the red dot. We can observe that most of landmarks are well located. The illustrated results of the validation set are better than the test set on ACXLDC dataset because of overfitting. The illustrated results of the test set on Huaxi-Analysis dataset are better than that on ACXLDC dataset.

#### 4.4. Case study

Fig. 12 illustrated some bad cases with large MREs. We consider three possible reasons why some landmarks were wrongly detected. First, some cephalometric images show complicated anatomic structures that render the landmarks blurry and some adjacent landmarks are affected by each other, such as (a, e, f, g, and h). Second, two copies of some landmarks were evidently visualized, such as (c). Third, the ear lever, teeth, or other structures cover and obscure some landmarks, such as (b and d).

### 5. Conclusion

In this paper, the two-stage framework for cephalometric landmark detection was developed. We employed the multi-task learning strategy to improve model generalization by aggregating Gaussian heatmap regression and offset map regression. We constructed the network with the *translational invariance limit* as the primary consideration. We modified three aspects of the two-stage framework: first, we reduced the number of downsampling layers in the LRN; second, we replaced the rest downsampling layers with anti-aliased pooling layers; and third, we proposed a blur-

ring upsampling module that shifts the feature maps and averages the upsampling results. The proposed method achieves robust performance on the private Huaxi-Analysis and public ACXLDC datasets.

In the real clinical environment, the results of the CLD method are robust across different datasets. We used the trained model on the ACXLDC dataset to detect the 19 shared landmarks on the Huaxi-Analysis dataset, and the MREs of the assigned landmarks' locations from the ground truth were  $1.77 \pm 0.82$  mm and  $1.76 \pm 0.92$  mm for the validation and testing sets, respectively. The corresponding results were  $1.92 \pm 1.36$  mm and  $2.22 \pm 1.94$  mm for validation and testing, respectively, on the ACXLDC dataset. Different X-ray capture devices can produce images with different properties of scale, brightness, or contrast ratio. Therefore, it is still a challenge to build a universal system. In our future work, we will increase the robustness of the CLD system across different datasets.

#### CRedit authorship contribution statement

**Tao He:** Conceptualization, Methodology, Writing - original draft, Visualization. **Jie Yao:** Software, Validation, Formal analysis, Data curation. **Weidong Tian:** Resources, Investigation. **Zhang Yi:** Supervision, Project administration. **Wei Tang:** Data curation, Validation. **Jixiang Guo:** Writing - review & editing.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgements

This work was supported by the National Major Science and Technology Projects of China under Grant No. 2018AAA0100201 and Regional Innovation Cooperation Project of Sichuan Province under Grant No. 2020YFQ0012.

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**Tao He** received a master's degree in computer science from Sichuan University, Chengdu, China, in 2017. Currently, he is a Ph.D. candidate at the College of Computer Science, Sichuan University. His research interests include artificial intelligence, medical imaging. He has published 9 academic papers. He volunteers to be the reviewer of international journals and conferences such as IEEE TCYB, IEEE TNNLS. He serves as a student member of IEEE and IEEE Computational Intelligence Society (CIS).



**Jie Yao** received a doctoral degree in Stomatology from West China School of Stomatology of Sichuan University, Chengdu, China, in 2020. She is a resident doctor of Stomatological Hospital of Xi'an Jiaotong University. Her current research interest is the application of Deep Learning in oral medicine.



**Weidong Tian** received a doctoral degree in clinical medicine from West China Medical University, Chengdu, China, in 1992. He is a Professor, chief physician, and the director of the Department of Traumatology and Plastic Surgery, West China Hospital of Stomatology, Sichuan University. He is the chief scientist of the National "973" Project and the National Key Research and Development Program. He is the director of the National Engineering Laboratory for Oral Regenerative Medicine and the Engineering Research Center of Oral Translational Medicine, Ministry of Education. He is the chairman of the Medical Research Management Committee of the Chinese Medical Association and the vice-chairman of the Medical Cosmetology Committee of the Chinese Medical Association. He is also the vice-chairman of the Plastic and Cosmetology Branch of the Chinese Medical Doctor Association and the vice-chairman of the Oral Biology Committee of the Chinese Stomatological Association. He has long been engaged in oral and maxillofacial trauma plastic surgery, medical cosmetology, stem cell, and regenerative medicine, and Cranial and maxillofacial digital technology. He has published 176 SCI articles and has been listed as "China's Most Cited Scholars" by Elsevier in the past 6 years.



**Zhang Yi** received a Ph.D. degree in mathematics from the Institute of Mathematics, The Chinese Academy of Science, Beijing, China, in 1994. Currently, he is a Professor at the College of Computer Science, Sichuan University, Chengdu, China. He is the co-author of three books: *Convergence Analysis of Recurrent Neural Networks* (Kluwer Academic Publisher, 2004), *Neural Networks: Computational Models and Applications* (Springer, 2007), and *Subspace Learning of Neural Networks* (CRC Press, 2010). He is a Fellow of IEEE. He is the Chair of IEEE Chengdu Section (2015 ~). He was an Associate Editor of IEEE Transactions on Neural Networks and Learning Systems (2009 ~2012), and an Associate Editor of IEEE Transactions on Cybernetics (2014 ~). He is also a Fellow of CAAL. His current research interests include Neural Networks and Big Data. He is the founding director of Machine Intelligence Laboratory. He is also the founder of IEEE Computational Intelligence Society, Chengdu Chapter.



**Wei Tang** received a doctoral degree in Stomatology from West China School of Stomatology of Sichuan University, Chengdu, China, in 2001. He is a Professor, doctoral tutor, and deputy director of the Department of Trauma and Plastic Surgery, West China Hospital of Stomatology, Sichuan University. He is the deputy director of the National and Local Joint Engineering Laboratory of Regenerative Stomatology, Sichuan University, an academician of the International College of Dentistry (FICD); He is the deputy head of the Academic Group of Head and Neck Surgery, Society of Reconstructive and Reconstructive Surgery, Chinese Association of Rehabilitation Medicine. He is the vice-chairman of the Society of Maxillofacial Surgery,



Sichuan Stomatological Association and Society of Stomatological Computer, Chinese Stomatological Association; He is the member of the Standing Committee of Branch of 3D Printing Technology, Chinese Medical Biotechnology Association, and the Expert Committee of 3D Printing Medical Application, Medical Management Center, National Health Commission of the People's Republic of China. He is also a member of the Society of Oral and Maxillofacial Surgery, the Chinese Stomatological Association. His current research interest is the application of Cloud Platform, Deep Learning, Augmented Reality, and Virtual Reality in oral medicine.

**Jixiang Guo** received a Ph.D. degree in the Department of Computer Science and Engineering from the Chinese University of Hong Kong, Shatin, NT, Hong Kong SAR, China, in 2011. Now, she is an associate professor with the Machine Intelligence Laboratory, College of Computer Science, Sichuan University, Chengdu, China. Her major research interest is the interdisciplinary area of artificial intelligence and healthcare, including deep neural networks, medical image processing, AI-assisted disease diagnosis, and preoperative planning.